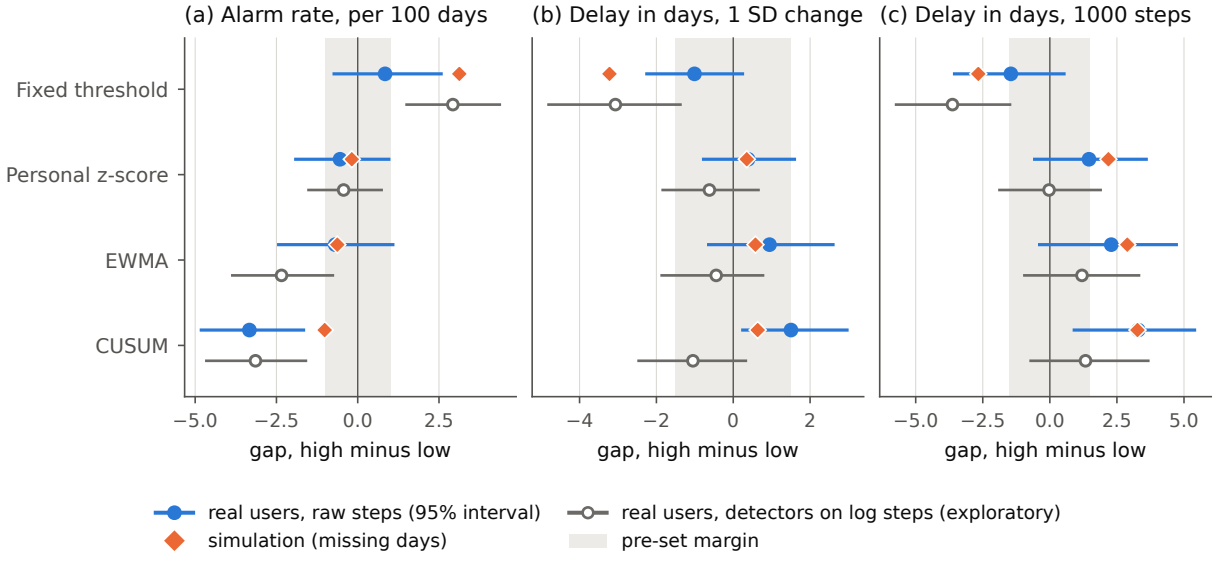
(a) Alarm rate, per 100 days
(b) Delay in days, 1 SD change
(c) Delay in days, 1000 steps
Fixed threshold
Personal z-score
EWMA
CUSUM
−5.0
−2.5
0.0
2.5
−4
−2
0
2
5.0
gap, high minus low
real users, raw steps (95% interval)
simulation (missing days)
real users, detectors on log steps (exploratory)
pre-set margin